Aiden Emanuel Fairman

Lugano, Switzerland • +41 79 489 39 59 • aidenfairman12@gmail.com
aidenfairman.vercel.app • github.com/aidenfairman12 • Nationality: Swiss & U.S.

EDUCATION

University of Bern — M.S. Bioinformatics & Computational Biology 09/2024 – 08/2026

Boston University — B.A. Computer Science 09/2019 – 08/2023

PROJECTS

Influenza Reassortment Detection via Transformer-based Contrastive Learning | github.com/aidenfairman12/DeepLearning

- Built a two-phase deep learning pipeline on large-scale genomic sequence data: a transformer encoder pre-trained with supervised contrastive learning (SupCon) feeding a downstream binary classification head.
- Iterated from an initial heterogeneous graph neural network (GATv2, multiple node types) to a transformer encoder better matched to the variable-length sequence data.
- Designed self-supervised augmentation (segment dropout, embedding noise) and on-the-fly synthetic example generation with auto-computed loss weighting to handle severe class imbalance.
- Built modular training infrastructure (early stopping, weighted sampling) with an evaluation suite reporting per-class AUROC and F1.

M.S. Thesis — Spatial Biomarker Validation in Colorectal Cancer (CosMx)

University of Bern, Institute of Tissue Medicine and Pathology (in progress, 2026)

- Built an end-to-end Python pipeline processing a 214-patient CosMx single-cell spatial transcriptomics cohort (6 TMA slides, 1,440 cores, 847,690 cells after QC), including Shapely-based polygon QC of per-cell boundaries.
- Ran Cox proportional-hazards survival analysis (lifelines, L2 regularization, Benjamini-Hochberg FDR) over the top 5,000 most variable supplied morphological and spatial features to test intra-TNM-stage-II prognostic stratification.
- Identified 56 significant multivariate features localizing to the tumor microenvironment and quantified cross-platform concordance against an independent GeoMx cohort.

ClimateQA: Multi-Model LLM Benchmark for Climate Science | github.com/aidenfairman12 | aidenfairman.vercel.app

- Built a 172-question multiple-choice benchmark grounded in IPCC AR6 passages, using four frontier LLMs (Claude, GPT-4o, Gemini, Llama) to generate and filter 580 candidate questions.
- Designed a self- vs. cross-model evaluation methodology that surfaced systematic biases in how models score questions they authored versus questions from other models.

NMR Metabolomics of Cardiac Graft Viability (semester project)

University of Bern

- Processed Bruker 1H-NMR spectra from 33 rat-heart perfusate samples (ischemia-reperfusion model) and built a Python multivariate pipeline (PCA via scikit-learn and OPLS-DA via pyopls) to detect ischemia-associated metabolite shifts, evaluating model fit by Q^2.
- Designed a dual-analyst reproducibility test across parallel processing pipelines to validate workflow robustness.

WORK EXPERIENCE

Novo Nordisk — Computational Biology Intern → Consultant 06/2025 – 06/2026

Boston, MA & Remote · part-time consultant (~8 hrs/week, via Kelly Services) following full-time internship

- Built a cross-species ventral tegmental area (VTA) single-nucleus atlas integrating >350,000 nuclei from ~20 rats, ~20 mice, and 6 human donors into a single reference object, delivered to Novo's internal h5ad/Seurat object platform for scientist use.

- Processed 4 single-cell / single-nucleus datasets (3 from published studies, 1 in-house) from raw Cell Ranger / Space Ranger outputs into analysis-ready objects (normalization, HVG selection, and batch integration) in both Scanpy and Seurat.
- Benchmarked Xenium vs. Visium spatial platforms on human adipose tissue to assess single-cell resolution; presented findings as the internship capstone.
- Supported GLP-1 and diet-response gene-expression research, translating analyses into decision-ready summaries for biologists and stakeholders in a regulated pharma R&D environment.

Center for Neurological Imaging — Brigham & Women's Hospital 03/2024 – 10/2024
Research Trainee, Harvard-affiliated laboratory, Boston, MA

- Prototyped a full-stack web module (JavaScript, Next.js) integrating Bruker MRI of MS-patient brain tissue with spatial transcriptomics from the same excised cores, as a proof-of-concept for a new data-exploration application.
- Embedded the Vitessce viewer for spatial-omics visualization and delivered the sample-display architecture, designing toward an MRI-ROI-driven core-selection workflow.
- Resolved conflicting Node.js and dependency versions across front end and back end to integrate the module within a larger multi-package platform.

MITRE — Software Engineer Intern 05/2022 – 08/2022
Bedford, MA

- Contributed to a 5-person Agile team that delivered the MVP of a C# mission-planning environment during the internship.
- Supported the transition of early prototypes toward stable, maintainable systems through structured development and documentation practices.

TECHNICAL SKILLS

Machine Learning: PyTorch, deep learning (classification, contrastive/SupCon learning), transformer encoders, graph neural networks (GATv2), survival analysis (Cox PH, lifelines), model training & evaluation, feature engineering

Bioinformatics: Single-cell & spatial transcriptomics (Scanpy, Seurat, Squidpy; 10x Chromium/Visium/Xenium, CosMx), Cell Ranger / Space Ranger, multi-modal data integration, QC pipeline design

Statistics & Analysis: PCA, OPLS-DA, dimensionality reduction, multivariate analysis, data visualization

Programming: Python, R, Bash, SQL, JavaScript, C#

Tools & Other: Git, full-stack web dev (Next.js, REST APIs), Vitessce, Agile/Scrum

LANGUAGES

English: Native • French: B1

ACTIVITIES & AWARDS

- **NCAA Division I Athlete**, Boston University Swim & Dive Team
- **Global Engineering Brigades**, BU chapter; clean-water infrastructure for under-resourced communities
- **Food Pantry Volunteer**, Boston Cares, Greater Boston Area